$$\int_{\sqrt{3}}^{9} \frac{\log_9 x}{x} dx$$

$$\frac{1}{\ln 9} \int_{\sqrt{3}}^{9} \frac{\ln x}{x} dx$$

$$u := \ln x$$

$$du = \frac{1}{x} dx$$

$$\frac{1}{\ln 9} \int_{\ln \sqrt{3}}^{\ln 9} u du$$

$$\frac{1}{\ln 9} \frac{1}{2} u^2 \Big|_{\ln \sqrt{3}}^{\ln 9}$$

$$\frac{1}{2 \ln 9} \left(\ln^2 9 - \ln^2 \sqrt{3} \right)$$

$$\frac{1}{2 \ln 9} \left(4 \ln^2 3 - \frac{1}{4} \ln^2 3 \right)$$

$$\frac{1}{4 \ln 3} \left(\frac{15}{4} \ln^2 3 \right)$$

$$\frac{15}{16} \ln 3$$